VEDRON STEM HLE DATA

Task Creation Guidelines

Graduate / PhD-level STEM problems that challenge frontier AI models

1. Overview

The objective is to assemble original, graduate-to-PhD-level STEM problems that are sufficiently challenging to cause today's most capable AI models to reason incorrectly, paired with a rigorous and accurate solution for each problem.

Each problem must have a single, exact, and independently verifiable answer. This dataset is used to train AI systems to reason dependably on advanced scientific problems — making both the difficulty of the question and the precision of the solution equally essential.

1.1 Domains in Scope

Domain	Accepted Subdomains
Biology	Molecular & structural biology, genetics & genomics, systems & computational biology, immunology, microbiology & virology, physiology, ecology & evolution.
Chemistry	Organic, inorganic, physical, biochemistry.
Physics	Quantum, electromagnetism, mechanics, thermodynamics & statistical mechanics, astrophysics & cosmology, condensed matter, nuclear & particle, optics.
Mathematics	Analysis, algebra, number theory, logic, geometry & topology, discrete mathematics & combinatorics, applied mathematics.
Engineering	Civil, mechanical, aerospace, chemical (mechanics/transport only), and computing topics (algorithms, ML, CV, NLP, robotics, networks, architecture).

Knowledge Cutoff: Every reference cited, and every fact your problem depends upon, must predate 31 August 2025. Do not cite any material published on or after that date.

2. How a Task Is Built

Every task follows the same eight steps. Sections 3 and 4 set out each requirement in detail; Section 7 provides the pre-submission checklist to complete before you submit.

1. Specify the domain and subdomain
2. Write the prompt — a single, self-contained, unambiguous question with exactly one correct answer
3. Provide the final answer and its format
4. Review responses generated by Opus 4.8 or GPT 5.4, where Pass@8 must be fewer than 4
5. Justify each failure with a precise Where / What / Why / Impact critique
6. Write the golden solution — your complete, original, step-by-step answer
7. Add open-access reference(s) predating 31 August 2025

8. Submit

3. Guidelines

3.1 Difficulty

Target graduate-to-PhD level. Every problem must demand deep, multi-step reasoning — not routine calculation or recall of a single fact.

Avoid textbook-style questions. Senior-level elective content may qualify; introductory material does not.

The aim is to write prompts that genuinely require reasoning. Each problem must be solvable by hand (pen, paper, basic calculator) by a domain expert within one hour. No code, Python, R, or spreadsheets.

3.2 Originality

Your submission must be entirely original — not reproduced from textbooks, papers, competitions, or websites, and not discoverable through a standard search.

References may serve as inspiration, but the problem itself must differ substantially. Changing only variable names or numerical values is insufficient.

Do not re-use any example problems in this document. Submit at most two problems that are substantially similar to one another.

3.3 Unambiguous

Ambiguity is the most frequent reason tasks are rejected. A prompt is unambiguous when any expert, relying solely on the text you supply, interprets it identically and reaches the same answer.

Exactly one question with exactly one final answer. No subjective or open-ended questions.

Where more than one valid method exists, all approaches must converge on the same answer.

Define every symbol, term, and abbreviation. Specify rounding for intermediate values and for the final answer. State units and the required answer format explicitly.

3.4 Self-Contained

The prompt must include everything required to reach the answer without any external lookup:

Exact values of all constants used.

All assumptions and conditions (reagents and equivalents, solvent, temperature, time, atmosphere).

Units for every quantity and any required data formats (e.g. SMILES, FASTA).

The specific model or equation to apply where more than one is plausible.

Reasoning vs. Knowledge: If the answer hinges on a fact that a model cannot reliably retrieve, a wrong answer represents a knowledge gap — not a reasoning failure — and does not qualify as a stump. Build problems whose difficulty lies in the reasoning, using information that is either provided in the prompt or constitutes common expert knowledge.

3.5 Requires Domain Expertise and Complex Reasoning

The problem must depend on specialised knowledge and several interconnected reasoning steps.

It must not be solvable by fact-recall alone. Strong problems combine multiple reasoning types (e.g. deduce a structure, then predict its behaviour under a modified condition).

3.6 Formatting (LaTeX / KaTeX)

The platform renders Markdown for text and KaTeX for mathematics. Verify that your maths renders correctly before submitting.

Use $ ... $ for inline maths and $$ ... $$ for display maths. Format Greek letters, symbols, and any non-trivial expressions in LaTeX.

Units must be non-italic and spaced from the number (e.g. $4.12\, \text{V}$). Broken LaTeX that confuses the models invalidates a failure.

3.7 The Final Answer — CURVD

Every answer must be Contained, Unambiguous, Reduced, Verifiable, and Discrete.

Requirement	Definition
Contained	Derived solely from the information provided in the prompt.
Unambiguous	Every expert arrives at the same answer.
Reduced	Expressed in the most concise form possible; no descriptive answers.
Verifiable	Every valid solution method yields the same answer.
Discrete	A single item: a number, expression, symbol/code, ordered list, name, or chemical formula.

Answer Formats

Format	When to Use
Integer	Whole-number counts (e.g. number of isomers).
Decimal	Precision results (constants, ΔG, rate constants, probabilities).
Fraction	Exact ratios or probabilities.
Text (case-sensitive)	Exact identifiers (chemical formula, IUPAC name, point group, gene symbol).
Text (case-insensitive)	Named concepts (reaction type, pathway, enzyme family).
Ordered / Unordered list	Sequences where order does, or does not, matter.

3.8 Not Allowed (Automatic Rejection)

Copied from or trivially adapted from existing sources, or answerable by a single search or database query.

Requires knowledge of events or publications dated on or after 31 August 2025, or contains personally identifiable information.

Open-ended or subjective, with no independently verifiable answer; or falling outside the approved domains.

Multiple choice — whether explicit or implied. Implied multiple choice includes yes/no, true/false, increase/decrease/no-change, and any question offering fewer than six plausible answers.

Multiple questions stacked into one; fill-in-the-blank tasks; or requests to "explain / describe" with no single discrete answer.

3.9 Guessability (Aim for 6+ Plausible Answers)

Design every problem such that a model would require at least six genuine attempts to land on the correct answer. A narrow answer space allows a model to guess correctly while reasoning incorrectly. Convert simple counts into ratios or multi-feature combinations to broaden the answer space.

4. Stumping Requirements

The bar every task must clear

1. Pass@8 < 4/8 on both Opus 4.8 and GPT 5.4
2. A valid failure requires BOTH a wrong final answer AND a genuine reasoning error

4.1 What Counts as a Valid Stump

Must Be True	Meaning
The final answer is wrong	It does not match the correct answer and differs meaningfully.
A genuine reasoning error caused it	Flawed logic, a misapplied principle or framework, an unjustified assumption, or a missing critical step.

Does NOT count as a stump:

Arithmetic or rounding mistakes alone.

Formatting issues, or correct reasoning accompanied by a minor numerical slip.

Misreading a sequence or string (e.g. a SMILES or DNA sequence).

Retrieving the wrong constant or value (a knowledge slip).

A correct final answer — no matter how flawed the reasoning, a right answer is never a stump.

Numerical Tolerance: Where responses fall within roughly 1–5% of the correct value, that is generally not a valid failure, even where the reasoning was flawed. Aim for wrong answers that diverge by at least ~5%; this can usually be arranged by selecting the problem's parameters accordingly.

4.2 Failure Justification (Required Structure)

For each response you mark as failing, write a precise critique. Refer to "the response" (not "the model") and name the specific response. Cover all four elements:

Element	What to Write
Where	The exact step or line where the error occurs.
What	The specific error (name the concept, theorem, or framework that was misapplied).
Why	Why it constitutes a reasoning failure and not a computational slip.
Impact	How the error leads to the wrong final answer, with numerical values where possible.

Good vs. Weak Justification

Good: "In Step 4 the response applies the ideal gas law to a high-pressure system where intermolecular forces are significant; it should have used the van der Waals equation because the density exceeds the ideal regime. This yields 2.1 atm instead of the correct 3.8 atm."

Weak: "The response arrived at the wrong answer because it made an error." — This is vague, names no reasoning failure, and points to no specific step.

5. Gold-Standard Examples

5.1 Chemistry — Cluster Symmetry Counting

Subdomain: Chemistry (Inorganic)

Prompt

> Consider a hypothetical compound formed from three closo-$C_2B_8H_{10}$ clusters joined by two B–B bonds (replacing hydrogens on the boron atoms). Assume that steric hindrance prevents two adjacent atoms within a cluster from both bearing bonds to other clusters, but that all other inter-cluster bonds are permissible. How many isomers could be formed in this manner, considering all possible closo-$C_2B_8H_{10}$ clusters and including both structural isomers and diastereomers, but counting enantiomeric pairs only once? Give the exact answer as an integer.

Final Answer: 79144

References: https://www.mdpi.com/2073-4352/10/10/896 (structure of $C_2B_8H_{10}$ clusters); https://math.mit.edu/~apost/courses/18.204_2018/Jenny_Jin_paper.pdf (overview of Burnside's lemma)

Failing Model Response

> Final answer given: 64,286
>
> Where it went wrong: At Step 4, the response divides by a symmetry group of order 4 only, having already mis-enumerated the cluster substitution positions.

Rationale for the Stump

> The response correctly identifies the bicapped square antiprismatic geometry of the closo-$C_2B_8H_{10}$ cluster, but fails to count the pairs of non-adjacent vertices accurately, obtaining 17 instead of 21. Furthermore, when enumerating the directed middle groups, it considers only C_2 symmetry as a proper rotation, ignoring C_2' and C_4 symmetry, thereby overcounting the directed middle groups. When classifying middle groups by symmetry, it considers individual symmetry elements only and does not account for clusters possessing multiple symmetries simultaneously (such as clusters with C_{2v} symmetry).

Golden Step-by-Step Solution

> Step 1: A closo-$C_2B_8H_{10}$ cluster has bicapped square antiprismatic geometry. The structure, before considering carbon placements and attachment points, has D_{4d} symmetry. With three clusters joined by two bonds, one cluster must serve as the 'central' unit bonded to two 'end' clusters. Because enantiomeric pairs are counted only once, structures are considered up to all symmetries including improper symmetries such as reflections.
>
> Step 2: Each possible end cluster has one attachment point and two carbon atoms. Burnside's lemma is applied to count the number of possible end clusters.
>
> Step 3: Counting all possible configurations, there are 10 ways to choose an attachment point and $C(9,2) = 36$ ways to choose two positions for the carbons, for a total of 360 configurations.
>
> Step 4: Configurations fixed by C_2 yield 2 ways to choose the attachment point and 4 ways to choose two carbons, for a total of 8 configurations.

Step 5: For each of the four sigma_d mirror planes, there are 4 ways to choose the attachment point and 6 ways to choose two carbons, for a total of 24 configurations.

Step 6: No end clusters are fixed by S8, S8^3, C4, or C2' symmetry elements.

Step 7: Applying Burnside's lemma, the total number of end clusters up to all symmetries is (360 + 8 + 4×24) / 16 = 29. Up to proper rotations only it is (360 + 8) / 8 = 46. There are 46 − 29 = 17 enantiomeric pairs, and 29 − 17 = 12 achiral end clusters.

Step 8: Treat each end cluster as a single unit and apply Burnside's lemma again to the central cluster. Two non-adjacent attachment points and two carbon atoms must be chosen.

Step 9: There are C(10,2) − 24 ways to choose two attachment points (excluding the 24 adjacent pairs), C(8,2) = 28 ways to choose two carbons, and 46^2 ways to choose two end clusters, giving 1,244,208 configurations in total.

Step 10: Configurations fixed by C2 total 9,200.

Step 11: Configurations fixed by each of the four C2' rotations total 552 each.

Step 12: Configurations fixed by each of the four sigma_d mirror planes total 2,672 each.

Step 13: No configurations are fixed by S8, S8^3, or C4.

Step 14: Applying Burnside's lemma, the total number of isomers (including structural isomers and diastereomers, counting enantiomeric pairs once) is (1,244,208 + 9,200 + 4×552 + 4×2,672) / 16 = 79,144.

Final Answer: 79144

5.2 Chemistry — Acid-Catalysed Epoxide Opening

Subdomain: Chemistry (Organic)

Prompt

C[C@]12[C@@H](CCC(C)=O)C(C)(C)CC[C@H]1O2 (4-((1S,2S,6R)-1,3,3-trimethyl-7-oxabicyclo[4.1.0]heptan-2-yl)butan-2-one, 1.0 g) was dissolved and stirred in dichloromethane (30.5 mL) at 0 °C. 0.45 mmol of p-toluenesulfonic acid was then added. The reaction was stirred for 12 hours at 0 °C, then a further 0.45 mmol of p-toluenesulfonic acid was added and stirring continued at 23 °C for 4 hours. The mixture was washed with saturated sodium bicarbonate (10 mL × 3), then dried over anhydrous magnesium sulfate. The solvent was removed and the product was purified by silica gel flash column chromatography. Compound X, the major product, was obtained in high yield as a colourless oil. Provide the identity of compound X as its Isomeric RDKit canonical SMILES string, including all relevant stereochemical descriptors.

Final Answer: C=C1[C@H](O)CCC(C)(C)[C@@H]1CCC(C)=O

Reference: https://doi.org/10.3390/md21040230

Failing Model Response

Final answer given: A bicyclic acetal (6,8-dioxabicyclo[3.2.1]octane framework)

Where it went wrong: The response invokes a "Cascade Cyclization" in the second step, asserting that "the ketone oxygen from the 3-oxobutyl side chain acts as an intramolecular nucleophile," then maps the resulting rigid bicyclic system to its final SMILES.

Rationale for the Stump

The response incorrectly posits a "Cascade Cyclization" in which the ketone oxygen acts as an intramolecular nucleophile. This does not occur and is chemically unsound: the actual transformation is a straightforward acid-catalysed epoxide ring-opening with elimination to form the exocyclic alkene — not an intramolecular cyclisation to a bicyclic acetal.

Golden Step-by-Step Solution

Step 1: Understand the chemical transformation. The overall reaction conditions facilitate an acid-catalysed ring-opening and alkene formation.

Step 2: The starting epoxide is treated with p-toluenesulfonic acid, which protonates the epoxide. The C1–O bond breaks to form a tertiary carbocation. The weak conjugate base then abstracts a proton from the C1 methyl group, forming an alkene and yielding compound X: 4-((1S,5R)-5-hydroxy-2,2-dimethyl-6-methylenecyclohexyl)butan-2-one.

Step 3: The isomeric RDKit canonical SMILES for compound X is C=C1[C@H](O)CCC(C)(C)[C@@H]1CCC(C)=O. Stereochemistry of the alcohol and side group is retained, as only the epoxide opens; that stereocentre disappears on alkene formation.

Final Answer: C=C1[C@H](O)CCC(C)(C)[C@@H]1CCC(C)=O

5.3 Chemistry — Photocatalytic Deposition (A Reasoning Trap)

Subdomain: Chemistry (Physical)

Prompt

An FTO/TiO2 electrode was introduced into a clear vial containing a 10 mM Pb2+ solution prepared in 0.1 M acetate buffer at pH 4.8. The electrode was irradiated with UV light at 1 cm for 5 min. After irradiation, the electrode was washed with ethanol and dried in air. XPS results show a high-resolution Pb 4f(7/2) peak at approximately 136.5 eV, and EDX mapping shows uniform distribution of the deposit. In the second step, the same sample was immersed in a 0.05 M CsBr methanol solution and kept in the dark for 30 minutes, then rinsed with 2-propanol, dried in air, and stored in the dark for further characterisation. What is the chemical formula of the deposited compound at the end of the process? Provide the answer in case-sensitive format (e.g. if the answer is iron, write Fe).

Final Answer: Pb

Reference: https://doi.org/10.1021/acsomega.2c03089 — step 1 photoreduction mechanism (p. 26741); step 2 dark stability (p. 26744)

Failing Model Response

Final answer given: CsPbBr3

Where it went wrong: The response writes an overall reaction "2 Pb + 6 CsBr + O2 + 2 H2O → 2 CsPbBr3 + 4 CsOH" and concludes the deposited compound is CsPbBr3.

Rationale for the Stump

In step 2, the response incorrectly assumes that chemical oxidation of metallic Pb back to Pb2+ is driven by dissolved oxygen within a 30-minute dark reaction. In practice, the sample is stable under these conditions and no appreciable reaction occurs. To form the perovskite in the second step, UV irradiation would be required to photo-oxidise Pb back to Pb2+ using the TiO2 surface as photocatalyst; only then would Pb2+ react with excess CsBr to form the all-inorganic perovskite. As conducted (in the dark), the deposit remains metallic Pb.

Golden Step-by-Step Solution

Step 1: The FTO/TiO_2 electrode is immersed in a Pb^{2+} solution and irradiated with UV light. TiO_2 photogenerates electron-hole pairs; the electrons reduce Pb^{2+} to metallic Pb while holes oxidise water. The XPS Pb 4f(7/2) peak at ~136.5 eV is characteristic of zero-valent metallic lead, confirming the deposit is Pb.

Step 2: The electrode is immersed in CsBr methanol solution in the dark for 30 minutes. The sample is stable under these conditions and no further reaction occurs. UV irradiation would be required to photo-oxidise Pb back to Pb^{2+}, which would then react with CsBr to form $CsPbBr_3$ — but no UV is applied.

Step 3: After step 1, metallic Pb is deposited. After step 2 (in the dark) the sample is unchanged. The deposited compound remains Pb.

Final Answer: Pb

5.4 Mathematics — A Constrained-Optimisation Supremum

Subdomain: Mathematics (Analysis)

Prompt

Find the supremum of the integral from 0 to 4 of f(x) dx, as f ranges over continuous functions [0,4] → R satisfying $x f(y) + 2y f(x) \le 3$ for all x, y in [0,4].

Final Answer: $(3/\sqrt{2}) \arcsin(1/\sqrt{3}) \approx 1.306$

References: None. References are optional outside Chemistry; this problem is self-contained.

Failing Model Response

Final answer given: 5/4 (= 1.25). Correct answer: $(3/\sqrt{2}) \arcsin(1/\sqrt{3}) \approx 1.306$

Where it went wrong: The response exhibits one admissible function whose integral equals 5/4, then asserts — without proof — that no continuous function can achieve a larger integral, treating a lower bound as though it were the supremum.

Rationale for the Stump

The proposed function has integral 5/4, but that is merely a lower bound for the supremum; the claim that "no continuous function can yield a strictly larger integral" is both unsupported and false. The response also assumes the maximiser satisfies f(4) = 0 without valid justification, and one of its derived bounds is wrong. Finally, it treats $H(x,y) = x f(y) + 2y f(x)$ as jointly concave — but H is not jointly concave on the relevant rectangles, and even a jointly concave function need not attain its maximum at corner points.

Golden Step-by-Step Solution

Step 1: Set $a = \sqrt{32/3}$ and $y(x) = \sqrt{16 - x^2/2}$. Since $y'(x) = -x / (2\, y(x))$, the function $y = y(x)$ satisfies the differential relation $x / (2\, y(x))\, dx = -dy$.

Step 2 (Upper bound): Substituting $y = y(x)$ into the constraint gives $x f(y(x)) + 2\, y(x) f(x) \le 3$ for all x in [0,a]. Dividing by $2y(x)$ and integrating over [0,a], a change of variables $u = y(x)$ converts the first term into an integral over [a,4], so the total integral of f over [0,4] is bounded by $M = (3/\sqrt{2}) \arcsin(1/\sqrt{3})$.

Step 3 (Attainability): The function g defined piecewise as $g(x) = (3/(32\sqrt{2}))\ \text{sqrt}(32 - x^2)$ for $0 \le x \le a$, and $g(x) = (3/(16\sqrt{2}))\ \text{sqrt}(16 - x^2)$ for $a \le x \le 4$, is continuous and satisfies the constraint everywhere (verified by Cauchy-Schwarz in each of the four regions).

Step 4 (Equality): For each x in [0,a], setting $y = q(x) = \text{sqrt}(16 - x^2/2)$ turns every inequality in Step 2 into an equality. Therefore the integral of g over [0,4] equals M, and the supremum is $M = (3/\sqrt{2})\ \arcsin(1/\sqrt{3})$.

Final Answer: $(3/\sqrt{2})\ \arcsin(1/\sqrt{3})$

5.5 Physics — Black-Hole Parker Wind (Astrophysics)

Subdomain: Physics (Astrophysics & Cosmology)

Prompt

Consider a stellar-mass black hole ($M = 10\ M\_sun$) in a High/Soft state radiating at a fixed fraction $\lambda = L/L\_Edd = 0.5$ of the standard Eddington luminosity, powered by accretion rate M_acc with radiative efficiency $\eta = 0.1$. The accretion flow and outflow comprise a fully ionised gas of primordial composition (75% H, 25% He by mass) at uniform Compton temperature $T\_C = 1.4 \times 10^7$ K. Determine the mean mass per particle (for the sound speed) and the mean mass per electron (for radiation interactions) from this composition.The thermal energy drives a steady-state, biconical, isothermal Parker wind subtending solid-angle fraction $f = \Omega/4\pi = 0.2$. The central radiation exerts an outward force on free electrons via Thomson scattering; account for how this modifies the effective gravitational potential. The wind accelerates through a sonic radius R_s; for integrated quantities assume the flow speed equals the isothermal sound speed c_s for all $r \ge R\_s$. The system self-regulates so that the integrated electron-scattering optical depth from R_s to infinity equals exactly unity.Constants: $m\_p = 1.673 \times 10^{-24}$ g; $k\_B = 1.381 \times 10^{-16}$ erg/K; $G = 6.674 \times 10^{-8}$ cm³ g^{-1} s^{-2}; $c = 2.998 \times 10^{10}$ cm/s; $\sigma\_T = 6.652 \times 10^{-25}$ cm².Determine the ratio M_wind / M_acc. Report a 2-significant-figure number; carry intermediate calculations to 6 significant figures.

Final Answer: 8.7

References: None. This problem is self-contained and built on standard isothermal Parker-wind and Eddington-luminosity theory.

Failing Model Response

Final answer given: 6.8. Correct answer: 8.7

Where it went wrong: The response computed the Eddington luminosity using the gas mixture's mean mass per electron μ_e instead of the pure-hydrogen proton mass, and then made a compensating error in the radiation-reduced gravity. The two errors partially cancel in the algebra and obscure the mistake until the final numerical result.

Rationale for the Stump

Step 2: The standard Eddington luminosity is defined for pure hydrogen (mass m_p), but the response computes it using the heavier mean mass per electron of the gas mixture (μ_e), inflating the luminosity and the accretion rate by a factor of 8/7.

Step 4: The response assumes radiation reduces gravity by exactly the fraction λ. In fact, radiation pushes outward on electrons while gravity acts on the full mass per electron (μ_e); since the luminosity is referenced to hydrogen but the gas is heavier, the correct reduction factor is $\lambda(m\_p / \mu\_e)$. Missing this makes the sonic radius too small.

Step 6: The term μ_e then cancels — but only because the Step 2 and Step 4 errors happen to offset each other. The helium content genuinely affects the dynamics; correcting both errors changes the ratio from the incorrect 6.8 to the correct 8.7.

Golden Step-by-Step Solution

Step 1 (Mean molecular weights): For $X = 0.75$, $Y = 0.25$, fully ionised: $1/\mu = 2X + (3/4)Y = 1.6875 \rightarrow \mu = 16/27 \approx 0.5926$; $1/\mu\_e = X + Y/2 = 0.875 \rightarrow \mu\_e = 8/7 \approx 1.1429$.

Step 2 (Isothermal sound speed): $c\_s = \sqrt{(k\_B T\_C / (\mu m\_p))} = \sqrt{[(1.381 \times 10^{-16} \times 1.4 \times 10^7) / (0.5926 \times 1.673 \times 10^{-24})]} = 4.416 \times 10^7$ cm/s.

Step 3 (Radiation-to-gravity ratio): The luminosity is referenced to the pure-hydrogen Eddington limit: $\Gamma = \lambda \mu\_e = 0.5 / 1.1429 \approx 0.4375$.

Step 4 (Sonic radius): $R\_s = G M (1 - \Gamma) / (2 c\_s^2) = G M (1 - \lambda \mu\_e) / (2 c\_s^2)$.

Step 5 (Mass outflow rate): From mass conservation and the $\tau = 1$ self-regulation condition: $M\_wind = 4\pi f \mu\_e m\_p c\_s R\_s / \sigma\_T = 2\pi f m\_p G M (\mu\_e - \lambda) / (c\_s \sigma\_T)$.

Step 6 (Mass accretion rate): $M\_acc = L / (\eta c^2) = \lambda L\_Edd / (\eta c^2) = 4\pi \lambda G M m\_p / (\eta c \sigma\_T)$.

Step 7 (Ratio): Dividing: $M\_wind / M\_acc = (f \eta) / (2\lambda) \times (c / c\_s) \times (\mu\_e - \lambda)$.

Step 8 (Final): $= 0.02 \times 678.9 \times 0.6429 = 8.728 \approx 8.7$ to two significant figures.

Final Answer: 8.7

6. Common Mistakes to Avoid

6.1 Multiple Choice + Ambiguous + Wrong Answer — Chemistry (Mass Spectrometry)

Subdomain: Chemistry (Physical)

Prompt (with errors): [A mass spectrometry prompt identifying an F2-isoprostane compound, asking for the name of the most polar isomer within a subclass showing the strongest intermolecular hydrogen bonding under normal-phase conditions, as an RDKit canonical SMILES string.]

What was wrong:

Implied multiple choice: the candidate isomers were enumerated in the prompt, allowing the model simply to select from the list.

Ambiguous: "most polar isomer / strongest hydrogen bonding under normal-phase conditions" is undefined — the column type, elution solvents, and gradient are unspecified.

Stacked question: it asks simultaneously for the compound class and the regioisomer.

Final answer wrong and unverifiable: the SMILES did not render (no backticks), did not match the stated formula, and the cited reference did not contain the spectrum of that isomer.

6.2 Under-Specified Conditions + Ambiguous Answer — Chemistry (Organic)

Subdomain: Chemistry (Organic)

Prompt (with errors): [A prompt describing diazotisation of an unspecified substituted aniline followed by azo coupling with possible azo-hydrazone tautomerisation, asking for the total number of geometrical (E/Z) isomeric elements in the final structure.]

What was wrong:

Not self-contained: no reagent identity ("a substituted aniline"), no equivalents, temperature, time, atmosphere, or solvent volume.

Ambiguous chemistry: "possible azo-hydrazone tautomerisation" leaves the structure undefined.

Ambiguous answer: "total number of geometrical (E/Z) isomeric elements" does not specify whether the count is per tautomer or across both, whether aromatic bonds count, or how the answer is to be interpreted. The format field also reads "Ans to this question ?".

No reference: every Chemistry task must cite an open-access source predating 2025.

6.3 Copied + Over-Length + Under-Constrained — Biology (Molecular & Cellular)

Subdomain: Biology (Molecular & Cellular)

Prompt (with errors): [A CRISPR/HDR prompt involving a frameshifted beta-lactamase gene and ampicillin resistance, with a nine-row allele table, asking what percentage of post-editing colonies carry allele P.]

What was wrong:

Final answer is incorrect: the intended answer must also count allele S, giving 17 / (17 + 14 + 8 + 7 + 8) = 31%.

Not novel: substantially a verbatim copy of a known training example.

Exceeds the length limit: the amino-acid sequences exceed the 100-character cap.

Under-constrained: the wild-type sequence is not provided, so it is impossible to determine which residues are mutations; it is also unspecified whether variation in the mature enzyme is tolerated, which affects the answer.

Missing rounding rules for intermediate values, and the stated final answer is incorrect (allele S omitted).

6.4 Guessable, Implied Multiple Choice — Chemistry (Organic)

Subdomain: Chemistry (Organic)

Prompt (with errors): [A Kulinkovich reaction prompt on a highly substituted cyclopentane, asking how many other substituents are in the same direction as the –OTMS group in the minor product.]

What was wrong:

Too few plausible answers: "how many other substituents are in the same direction" has well under six possible outcomes, making it effectively a multiple-choice question.

Ambiguous wording: "substituent" is undefined (does hydrogen count? must it be on a specific ring?), and "same direction" is not anchored to a reference frame.

Incomplete conditions for a Kulinkovich reaction: the required air-free environment and the workup procedure are absent.

6.5 Invalid Stump + Significant-Figure Mismatch — Physics (Condensed Matter)

Subdomain: Physics (Condensed Matter)

Prompt (with errors): [A graphene/n-type silicon Schottky junction prompt asking for current density under a bias voltage, with extensive material parameters, requesting a 3-significant-figure answer.]

What was wrong:

Invalid stump: the responses that "failed" did so through unit-conversion and arithmetic errors, not reasoning errors. A stump requires a genuine reasoning error in at least 3 of the 6 responses.

Significant-figure mismatch: the answer is requested to 3 significant figures, but several given parameters carry only 1–2 (e.g. N_D = 2.0 × 10^15). Every value must carry at least as many significant figures as the required answer.

Unstated quantity: the Fermi velocity of graphene is not a fundamental constant and must be supplied explicitly; omitting it allows different solvers to arrive at different answers.

A reviewer also questioned aspects of the golden solution's physics (which energy bands to include) — a reminder that your solution must itself be airtight, not merely the prompt.

6.6 Undefined Term + Not Novel — Mathematics (Graph Theory)

Subdomain: Mathematics (Discrete / Graph Theory)

Prompt (with errors): Compute the radio number of the corona graph P2 ⊙ W31, where P2 is the path on two vertices and W31 is the wheel on 31 vertices.

What was wrong:

Not self-contained: "radio number" is not defined, and the two standard conventions yield different answers — the positive-label convention gives 67, the non-negative/span convention gives 66. A prompt admitting two correct answers is inherently ambiguous.

Internally inconsistent: the supplied solution begins labelling from 0 (the non-negative convention) but concludes with 67 (the positive convention).

Not novel: the result follows mechanically from applying published theorems in the cited paper; the golden solution adds no original reasoning. Original reasoning beyond simply citing a published result is required.

7. Pre-Submission Checklist

Complete this checklist before submitting. Any unchecked item is a likely grounds for rejection.

Prompt

☐	The prompt is fully self-contained.
☐	All variables, constants, assumptions, and definitions are stated.
☐	The prompt is unambiguous: one question, one answer, no hidden assumptions.
☐	Solvable by hand within one hour; no internet access or external tools required.
☐	Not multiple choice (explicit or implied); at least 6 plausible answers.
☐	Graduate / PhD level; requires multi-step reasoning, not formula recall.
☐	Problem is 100% original.
☐	Prompt is not reproduced from textbooks, papers, competitions, or websites.
☐	Prompt is not discoverable through an online search.
☐	Prompt is not a trivial modification (variable/name/number changes only) of any existing source.
☐	Prompt clearly states the required output format.
☐	Mathematical notation renders correctly in Markdown.

☐	No Markdown-breaking formatting errors; units formatted correctly (especially Physics and Chemistry).

Final Answer

☐	Answer is unique and unambiguous.
☐	Answer is independently verifiable.
☐	Answer matches the solution exactly.
☐	Answer has the correct number of significant figures (if specified).
☐	Answer has correct units (if applicable).
☐	Correct answer format selected.

Model Failures and Justification

☐	Pass@8 < 4/8 (fewer than 4 of 8 attempts correct) on Opus 4.8 or GPT 5.4.
☐	Each justification covers Where / What / Why / Impact, names the specific response, and uses the phrasing "the response".
☐	Failures are genuine reasoning errors (not arithmetic, formatting, parsing, or knowledge slips).
☐	Explanation clearly articulates why the reasoning is incorrect.
☐	The specific step where the failure occurs is referenced.
☐	Reasoning error distinguished from calculation error.
☐	Explanation is precise and technical (not vague).

Golden Solution and References

☐	Every step is explicit; all intermediate maths is shown; all assumptions and theorems are justified.
☐	Every reasoning step is written out explicitly.
☐	All intermediate calculations are shown.
☐	All formulae are justified.
☐	All assumptions are stated.
☐	No "it is obvious" shortcuts.
☐	Steps are numbered; the final answer is clearly labelled and in the required format.
☐	A reviewer can independently verify the solution.
☐	All algebra verified.
☐	All constants checked.
☐	Units are consistent throughout.
☐	Final answer matches the derivation and is clearly labelled.
☐	Reference(s) are open-access and predate 31 August 2025, and actually support the problem.